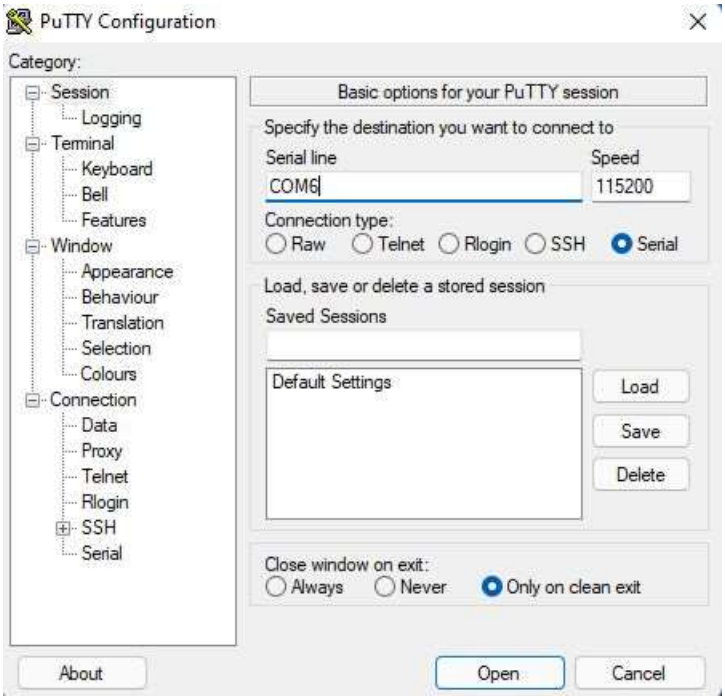
EBAZ-4205 Arcade Linux OS Configuration

- EBAZ-4205 ZYNQ-7010 FPGA Board
- USB to Serial Adapter
- Putty Software
- PinballWiz EBAZ4205-Arcade_v15 GitHub Repository Files

USB to Serial Adapter (aliexpress.com)

Putty Software



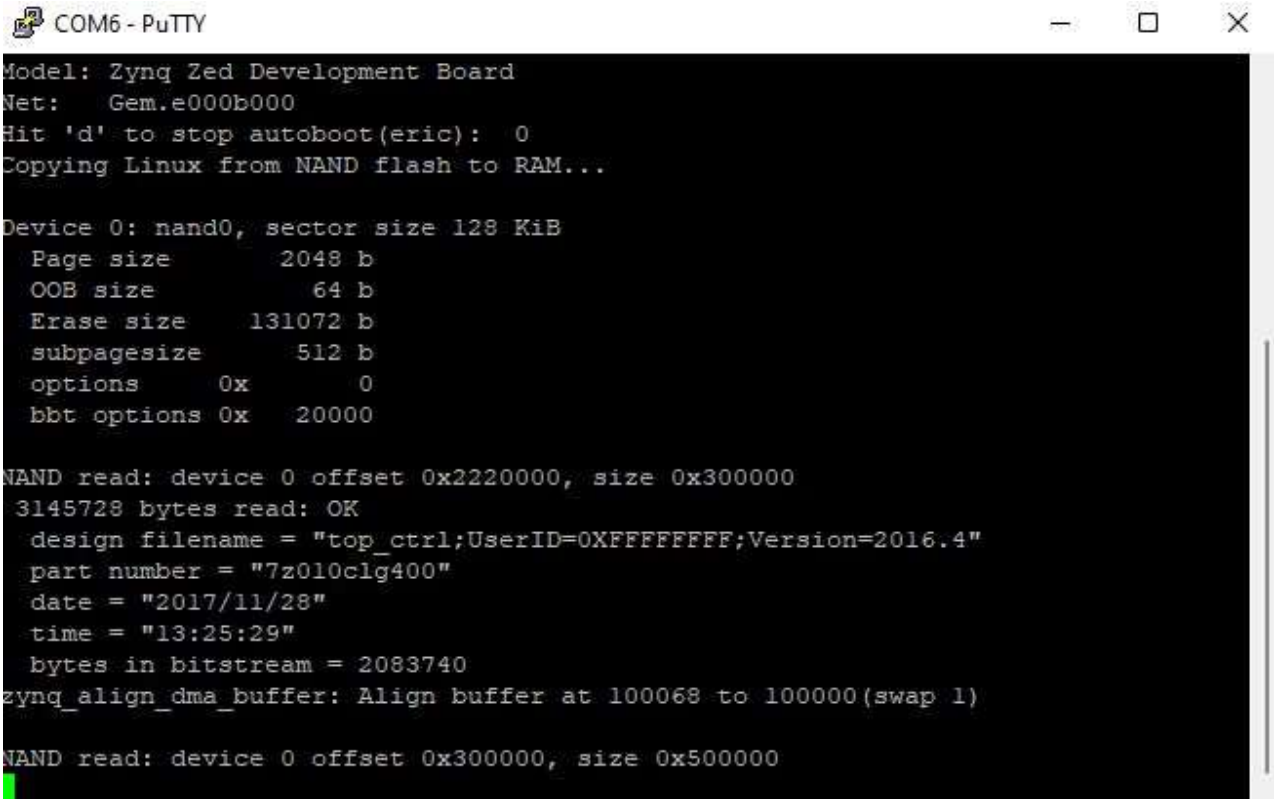
Serial Communication Setup

Connect Serial Adapter Gnd, RX, TX to EBAZ-4205 Gnd, TX, RX pins on serial header as shown below (VCC is not used). Note RX to TX and TX to RX connections below. Connect Serial Adapter to USB Port on Computer. Open Putty Software and setup as shown above:

- Select Serial
- COM# (device manager will show which com port to use)
- Speed = 115200

Click Putty Open Button.
Power On the EBAZ-4205 Board.

Putty Should now display the EBAZ-4205 Boot process.



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- Interrupt the normal boot process in Putty by pressing 'd' key.
- The Putty Console Window should then show the prompt below.
- The root password can now be changed. (see commands below).

COM6

```
gethostbyname: Host name lookup failure0.000 (avg):0.000h/s
their_ipaddr =
Usage: e.g703.cn hostname
gethostbyname: Host name lookup failure0.000 (avg):0.000h/s
their_ipaddr =
Usage: e.g703.cn hostname
═══~═══════(1m):0.000 (5m):0.000 (15m):0.000 (avg):0.000h/s

U-Boot 2015.07 (Jun 28 2017 - 21:05:18 +0800)

Model: Zynq Zed Development Board
DRAM:  ECC disabled 256 MiB
NAND:  128 MiB
MMC:   zynq_sdhci: 0
Invalid bus 0 (err=-1)
*** Warning - spi_flash_probe() failed, using default environment

In:    serial
Out:   serial
Err:   serial
Model: Zynq Zed Development Board
Net:   Gem.e000b000
Hit 'd' to stop autoboot(eric): 0
zynq-uboot> _
```

Resetting Root Password

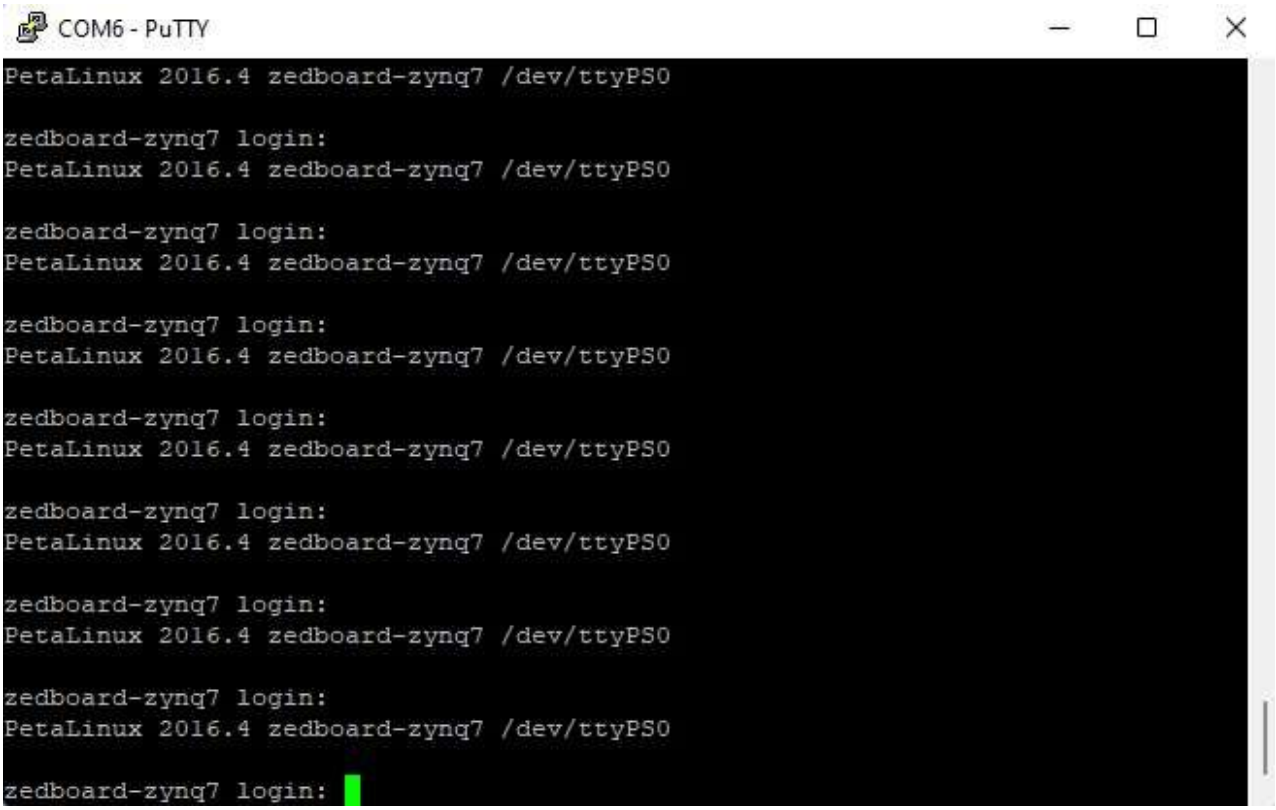
Copy each of these command lines into the Putty Console window - hit enter after each.

```
nand read ${kernel_load_address} 0x300000 ${kernel_size}

nand read ${devicetree_load_address} 0x800000 ${devicetree_size}

setenv bootargs 'console=ttyPS0,115200 root=/dev/mtdblock6 rootfstype=jffs2 noinitrd rw rootwait reboot=cold,hard emergency init=/bin/sh'

bootm ${kernel_load_address} - ${devicetree_load_address}

passwd
```

- Choose a root password when prompted and verify password again.
- Restart the EBAZ-4205 Board.

Disable Bitcoin Miner Software

- Apply Power to the EBAZ-4205 Arcade Board.
- The EBAZ-4205 Board should eventually show the below login prompt.
- Login as root using the password set in the previous steps.
- The EBAZ-4205 Board should also still be running the Bitcoin Miner Software which will continue to display information in the Putty Console Window.



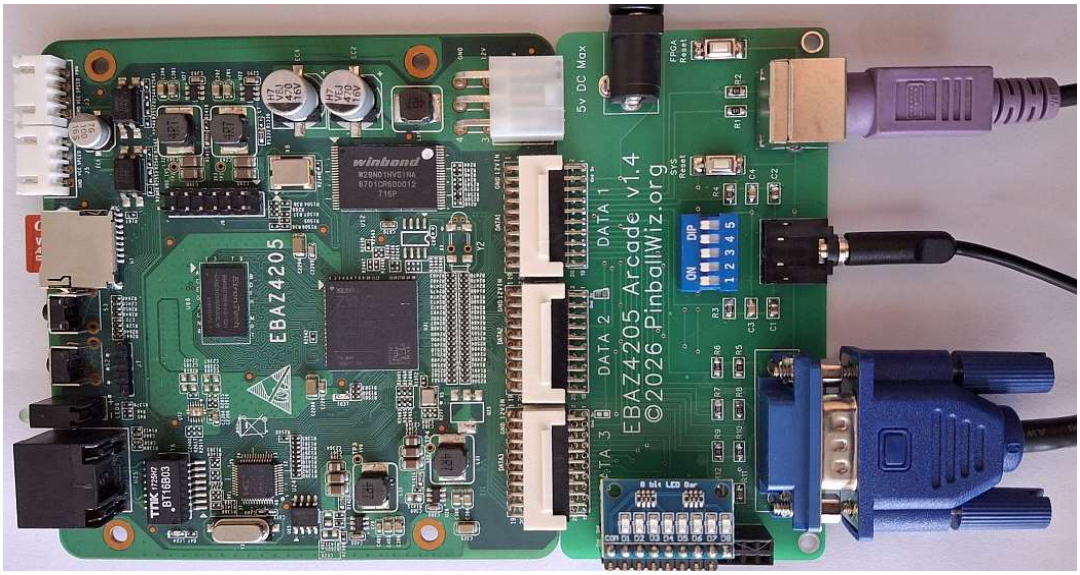
- To Disable the Bitcoin Miner Software Enter the commands shown below.
- Hit Enter after each command.

```
mv /etc/rcS.d/S95cgminer.sh /etc/rcS.d/K95cgminer.sh

reboot
```

Copying EBAZ-4205 Arcade Files to Linux OS

- Format a 2GB or larger MicroSD Card as FAT32
- Copy the Linux folder from the EBAZ4205-Arcade_v15 Repo to the MicroSD Card.
- Place the MicroSD Card into the EBAZ-4205 MicroSD Card Slot.
- Copy the following Commands into the Putty Window hitting Enter after each.

```
mkdir -p /mnt/sdcard

mount -t vfat /dev/mmcblk0p1 /mnt/sdcard

cp /mnt/sdcard/linux/etc/init.d/fpga_loader.sh /etc/init.d/fpga_loader.sh

cp /mnt/sdcard/linux/var/dipread.bit /var/dipread.bit

cp /mnt/sdcard/linux/var/game_loader /var/game_loader

ln -s /etc/init.d/fpga_loader.sh /etc/rcS.d/S96fpga_loader.sh

shutdown -h now
```

- Linux OS Configuration is now complete.
- Remove power from the EBAZ-4205 Board.
- See included Documentation on the MicroSD Card Games Setup procedure.

Once a MicroSD Card has been setup and loaded with games, Insert the MicroSD Card into the EBAZ-4205 MicroSD Card slot. Set the Dipswitch according to the game you wish to load (see EBAZ-4205 Arcade GameMap doc for Dipswitch Settings). Connect a VGA Monitor, Audio Speakers and PS2 Keyboard and apply power to the EBAZ-4205 Arcade Board. The Game selected should load automatically.